Transient Stability Analysis in Power Systems Integrated with a Doubly-Fed Induction Generator Wind Farm

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***Abstract*—Renewable energy systems, especially wind turbines and farms are nowadays integrated rapidly into power systems and smart grids. Several technical challenges arise due to penetration of wind energy into power networks and systems. In order for system stability and steady state operation to be ensured in power systems and electric networks, steady and dynamic analysis are necessary. We study a standard IEEE 9 bus test system, which is integrated with a Doubly-Fed Induction Generator wind farm, aiming to examine its behaviour during disturbances. Steady state and transient stability configurations are proposed in order to analyze the system described. A three-phase fault is suddenly applied to a load bus. Moreover, critical fault clearing time is identified for the corresponding three-phase fault and results show the maximum time for system stability during a disturbance. The influence of transient stability, including voltage stability, angle stability, active power and reactive power is discussed and research results become important for the smooth integration of wind farms into networks. This study is conducted with the help of PSS/E 34 software simulation tool by Siemens.**

***Index Terms*—Renewable energy sources; Doubly-Fed Induction Generator; Steady State Analysis; Transient Stability Analysis; Critical fault clearing time;**

I. Introduction

Nowadays renewable energy sources have invaded into modern, traditional and smart grids. The depletion of fossil fuels, environmental concerns and health implications on humans have turned societies to the use of renewable energy sources. Wind power is a widespread renewable energy technology and like all renewable energy sources its integration into grids presents some difficulties and challenges. The unpredictable nature and its volatility are the most prominent problems of wind energy. This variable nature of wind power causes voltage stability and small signal stability problems, oscillations in frequency and the distribution of power flow is influenced. The transient stability of networks and power system integrated with large-scale wind power is a hot issue and many studies have focused on it.

In order for a system to be secure and stable, it should be able to maintain a steady condition under major disturbances like faults in lines or buses and sudden loss of equipment. Steady-state analysis is directly linked with the calculation of load flow on the voltage profiles of system bus bars, transmission lines and transformers and plays an important role for the examination of the stability of a power system integrated with wind power [1]. Furthermore transient stability is the capability of the power system to maintain synchronism under severe transient disturbances and is crucial for the stability and smooth operation of the system [2]. Doubly-fed induction Generator (DFIG) wind turbines are used in wind farms to a large extent and variable speed turbine generators are becoming more popular in energy markets. The aim of this study is to investigate the integration of a Doubly Fed Induction Generation Wind farm in a power system, especially in relation to the stability of the system during a three-phase fault, and to determine critical fault clearing time for the system. In this paper a standard IEEE 9-bus system is chosen as a benchmark for the simulations and the wind farm integration is done directly through a transformer without the use of any shunt capacitors or STATCOM. The results obtained of the simulation of this paper could offer some operational references for practical power systems integrated with large scale DFIG wind farms. In addition, the transient results confirm the smooth integration of wind energy in power systems as the time the system recovers to its initial condition is very small. This paper is divided into five sections. Section II analyzes related work; Section III contains the description of the system; Section IV presents the modeling of the power system and especially the Doubly Fed Induction Wind farm; In Section V the simulation results are analyzed and discussed and finally in Section VI the main points of the study and the main conclusions are highlighted.

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II. Literature Review

Interconnection of wind energy sources into large and complex electrical networks and power systems is a subject with growing interest. Prior work and studies have been performed in transient stability analysis in relation to integrating wind turbines into grids. Various wind turbines are selected for transient stability analysis. Firstly, [1] examines the integration of large scale wind farms with Doubly fed induction generator wind turbine into the Moroccan southern transmission network through steady-state and dynamic analysis simulations. In [2] various system faults have been simulated in order for the impact of wind turbines connected to the Australian grid to be studied. PSS/E software is used in [3] in order for the transient simulation results of an offshore wind farm connected to Taiwan Power System to be obtained. The transient response simulation results of the Taiwan Power System connected with a large-scale offshore wind farm of 100 MW subjected to symmetrical and unsymmetrical disturbances are examined in [4]. [5] analyzes the influences of transient stability, including voltage stability, angle stability and frequency stability of large scale wind power accessing in regional grid. In [6] the effects of integrating a wind power plant on a single machine infinite bus test system are highlighted. Moreover, the increase of the wind penetration and its effect on the critical clearing time of the system is studied. Wind power integration during transient stability phenomena are analyzed in [7] using EUROSTAG software. Additionally, [8] discusses the IEEE 14 bus test system under a particular loading and fault condition, examining the integration of wind with the existing power system, [9] examines as well the wind integration in IEEE 14 bus test system and identifies weak nodes for the use of shunt capacitors and [10] uses PSAT software for power flow, continuous power flow and transient stability analysis in the same bus system. Furthermore, [11] uses transient stability analysis in order to analyze the system stability, the active and reactive power response, the voltage response-fault ride through ability when wind farms are connected into the Kosovo transmission grid. In contrast to [11], [12] studies the impact of integration of wind farms of full scale converter wind turbine with permanent magnet synchronous generator in Kosovo power system. In [13], wind generators and solar photovoltaic systems, on transient stability of synchronous generators connected to IEEE 9 bus system are discussed. Mathematical modelling for the study of the steady state and transient analysis on wind energy system is presented in [14]. In [15] the transient voltage stability of offshore wind farms following faults on the offshore collector system are presented. Finally [16] and [17] test the IEEE 9 bus system which is chosen as a benchmark for this study, without the integration of renewables. This study complements this work and aims to conduct a transient stability analysis of the IEEE 9 bus system when it is connected directly to the grid through a transformer with a wind farm of 51 MW of Doubly Fed Induction Generator wind turbines. A three-phase fault is applied on a load bus and the system's behaviour is interpreted.

In general, our paper succeeds in a wind power penetration of almost 16%. More specifically, compared to studies, [3], [9], [11], our study as mentioned earlier has a greater merit of renewable penetration (16% compared to renewable penetration of the studies mentioned, which present under 10% of renewable integration) and presents a more stable behaviour under fault conditions. Transient stability results that are presented in Section V, confirm that our network integrated with a DFIG wind farm can withstand a three-phase fault up to 0.75 sec (750 msec). Integration of wind power into the IEEE 9 bus system is smooth and can respond and withstand disturbances as the stability and operation of our network are ensured.

III. Description of the electrical network

A. IEEE 9 bus system

The IEEE 9-bus system is a common test system template for power flow and dynamic simulation studies by engineers and researchers in order to study stability of the system and simulate new algorithms and techniques. The IEEE 9-bus system is diagrammed and simulated with the PSS/E software by Siemens. This system is composed of 3 loads, 3 transformers and 3 generators/machines, which are interconnected through a network of 9 buses. Total generation is of 319.6 MW and the total load which is supplied by the generators is of 315 MW of active power and 115 MVAR of reactive power. Swing bus is considered for this study bus 1.

B. IEEE 9 bus system with Wind Farm

In this study, a wind farm of 51 MW (approximately 16% of the total generation) is added to the standard IEEE 9 bus system. The wind farm is composed of machines of the same type DFIG (Doubly-fed induction generator) of 1.5 MW. As a result of this an equivalent machine of the wind farm output power is used in order to model the whole wind farm. The Wind farm (bus 10) is connected to the grid (bus 7) through a transmission line of 230 kV and between these buses there is a 13.8/230 kV step up transformer.

IV. Modeling of the system

A. Steady state analysis modeling

Steady state modeling is considered to be a very simple process in the PSS/E software. It mainly concerns the modeling of various parts of the system like cables, transformers and generators. These elements have to be initialized with correct parameters according to the parameters of the whole network. The most important part of steady state analysis is the load flow solution. Load flow solution provides the power flows on transformers, transmission lines and buses and plays a major role for the smooth connection of wind farm to the network. The calculation of power flow as well, provides the initial condition for the dynamic simulations.

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B. Dynamic model of wind farm and the rest network

The PSS/E software was developed for simulations with various generator models. PSS/E supports 4 types of wind turbines as follows [18]:

- Type 1: Direct connected Conventional Induction Generator.
- Type 2: Wound rotor Induction Generator with Variable Rotor Resistance.
- Type 3: Doubly-Fed Induction Generator DFIG.
- Type 4 Full Size Converter Unit.

In general Generic Wind Models are used in studies concerning integration of renewables in a power system. Nowadays variable speed wind turbines are considered to be a widespread technology. Type 3 and type 4 are variable speed wind turbines which provide solutions to problems fixed speed wind turbines face, like fluctuations in voltage and power generated into the grid. A device allows rotation speeds variation in variable speed wind turbines; thus kinetic energy is stored and as a result of this power fluctuations are mitigated [19]. A Doubly Fed Induction Generator wind turbine is selected for this study and it can be seen in Fig. 1.

As mentioned earlier the second generation of Doubly Fed Induction Generator wind turbine (Type 3 Generic model) in PSS/E is used for the dynamic model of the wind turbine. The generic WT3 model of second generation is included in version 34 of PSS/E. The generator in the power flow data file has to be implemented as a wind machine in order to use the WT3 model. In Fig. 2 the structure of second generation type 3 wind turbine generator (WTG) model is presented. The model consists of seven modules [20]:

1) The renewable energy generator/converter ($regc_a$) model, which has inputs of real ($Ipcmd$) and reactive ($Iqcmd$) current command and outputs of real (Ip) and reactive (Iq) current injection into the grid model.
2) The renewable energy electrical controls ($reec_a$) model, which has inputs of real power reference ($Pref$) that can be externally controlled, reactive power reference ($Qref$) that can be externally controlled and feedback of the reactive power generated ($Qgen$). The outputs of this model are the real ($Ipcmd$) and reactive ($Iqcmd$) current command.
3) The emulation of the wind turbine generator drive-train ($wtgt_a$) for simulating torsional oscillations. The output of this model is speed (spd). In this case speed is assumed to be a vector spd = [ωt ωg], where ωt is the turbine speed and ωg the generator speed. The mechanical power/torque can be varied through the action of the connected aero-dynamic model.
4) A simple model of the wind turbine generator aerodynamics ($wtgar_a$).
5) A simplified representation of the wind turbine generator pitch-controller ($wtgpt_a$). This is similar to the existing type 3 pitch-control model, with the addition of one parameter Kcc.
6) A simple emulation of the wind turbine generator torque control ($wtgtrq_a$).
7) A simple renewable energy plant controller ($repc_a$), which has inputs of either voltage reference (V_{ref}) and measured/regulated voltage (V_{reg}) at the plant level, or reactive power reference (Q_{ref}) and measured reactive power (Q_{gen}) at the plant level. The output of the $repc_a$ model is a reactive power command that connects to Q_{ref} on the $reec_a$ model [19].

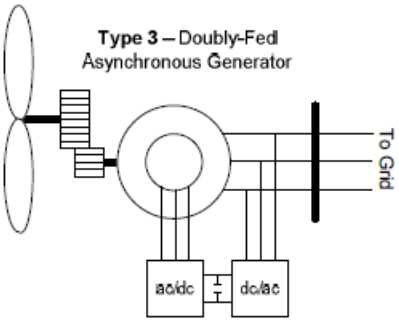


Fig. 1: Type 3 - Doubly fed Induction Generator [20].

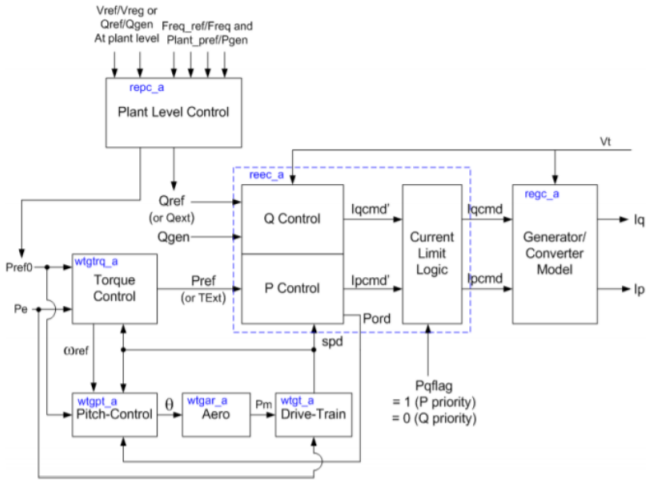


Fig. 2: Type 3 WTG model [20]

V. Results and Discussion

A. Load flow analysis

As mentioned earlier load flow analysis plays an important role for the power system and for the dynamic simulation as well. Different methods are proposed for load flow solutions, and amongst them two major ones are Gauss-Siedel and Newton-Raphson. The complexity of modelling and the type of study determine which method is suitable for the load flow analysis. The PSS/E offers an additional method, namely the Fixed slope decoupled Newton-Raphson, which is the one chosen for the load flow analysis in this study. The load flow can be seen in Tables I and II.

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TABLE I: Load Flow in all buses of the system

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)	Angle (deg)
1	BUS1	16.5	1	AREA1	1.04	0
2	BUS2	18	2	AREA2	1.025	15.47
3	BUS3	13.8	2	AREA2	1.025	8.91
4	BUS4	230	1	AREA1	1.0187	-0.74
5	BUS5	230	1	AREA1	0.9827	-1.03
6	BUS6	230	1	AREA1	1.0048	-1.28
7	BUS7	230	2	AREA2	1.016	9.85
8	BUS8	230	2	AREA2	1.008	6.03
9	BUS9	230	2	AREA2	1.0286	6.2
10	WIND_FARM	13.8	1	AREA1	1.0058	11.53

TABLE II: Active and reactive power of the machines of the system

Bus Number	Bus Name	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
1	BUS1 16.500	23.8063	250	10	38.6783	300	-300
2	BUS2 18.000	163	300	10	22.8221	300	-300
3	BUS3 13.800	85	270	10	-4.2249	300	-300
10	WIND_FARM 13.800	51	52	13.25	-16.7629	16.7629	-16.7629

B. Transient Stability Analysis

The dynamic analysis examines the condition of the system, right after the three-phase fault or any disturbance applied to a line near a bus in the network. Moreover, it detects the critical fault clearing time (CFCT); this is defined as the maximum time during which a disturbance can occur without losing system stability [1]. The calculation of this specific time confirms the stable operation of the system during the various disturbances or shocks it may be subjected to. The main aim of this study is to ensure that the integration of renewable energy technology in a system will not affect the system stability and its smooth operation. Additionally, a reasonable short critical fault clearing time is sought, in order for the system to reset in its initial state of operation.

For the transient stability analysis in this study, a three-phase fault is applied to line 4-5 near bus 5 which is a load bus in a test system. The analysis is performed in the following steps:

1) The system is initialized.
2) The fault is applied one second after the initiation of simulation in a bus.
3) The fault duration increases gradually in steps of 50 msec until the system becomes unstable.
4) Then the fault is cleared by opening line 4-5.
5) The simulation continues until reclosure of the tripped line.

In this system, the CFCT (critical fault clearing time) was found to be 0.75 sec (750 msec). The total simulation time was 10 seconds and the fault occurred in bus 5. After 0.75 sec the fault was applied, the three-phase fault was cleared and the line 5 to 4 was tripped. The simulation continued for 1 second and after the line 5 to 4 was closed. After 0.75 seconds, a phase synchronization is observed between the relative with respect to generator 1 power angles of generators 2 and 3. The system is stable for the fault duration of 0.75 sec and it can be seen in Fig. 3 where the relative power angles of generator 2 and 3 with respect to generator 1 are presented. If the fault duration increases to 0.80 sec then the system becomes unstable.

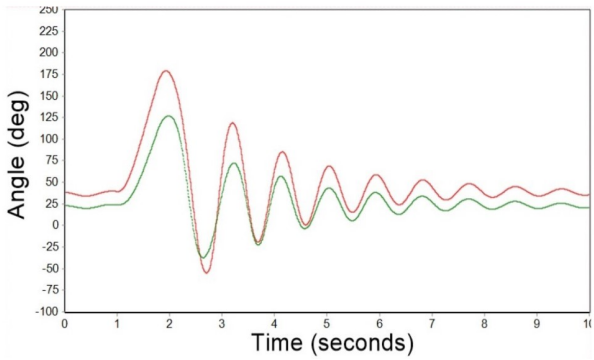


Fig. 3: Power angles of generators 2 and 3.

Fig. 4 illustrates the active power of the wind farm. It can be easily noticed from this figure that the active power of the wind farm decreases nearly to zero after the fault occurs. When the fault is cleared the active power of the wind farm recovers quickly enough to the starting value of 0.3 pu and stays stable to this value but with the existence of some oscillations. Active power responses may be explained by the acceleration of the rotor and as a result of this a larger negative slip during the fault. Theses oscillations may be due to the model of the wind turbine. In Fig. 5 the active power of all machines of the system can be seen. After the fault there is a decrease in active power of all machines and then oscillations with great width for the conventional machines are observed.

Reactive powers of the machines of the system are illustrated in Fig. 6. Before the occurrence of the fault the reactive power is seen to be stable at -0.168 pu (-16.8 MVAR). After the fault wind farm reactive power presents oscillations between values of -0.40 pu to almost 0.1 pu. Great changes in the value of reactive power can be seen as well for the conventional machines.

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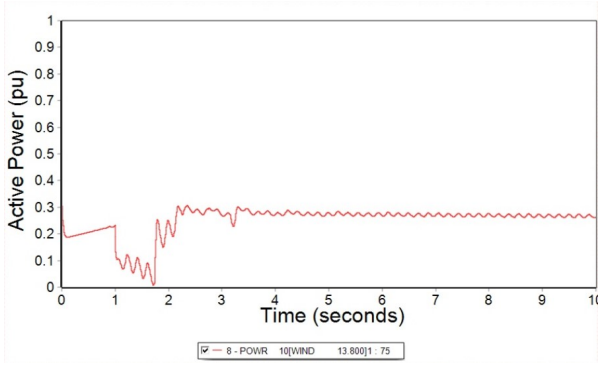


Fig. 4: Active power of Wind Farm for CFCT 0.75 sec.

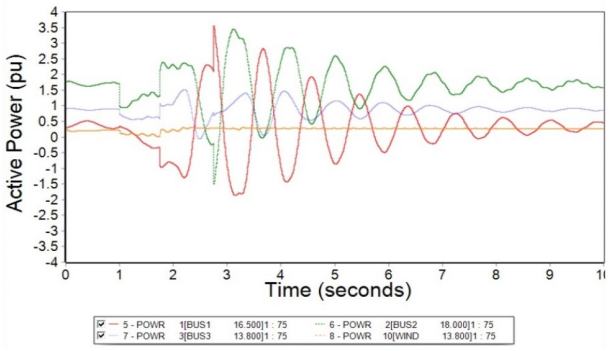


Fig. 5: Active power of machines for CFCT 0.75 sec.

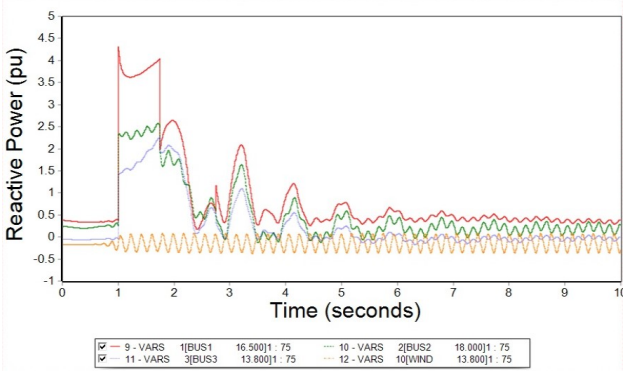


Fig. 6: Reactive power of all machines of the system.

Right after the three-phase fault in bus 5, a sudden voltage drop is observed in all buses of the system. The drop only lasts for a very short time so voltage of all buses gradually recovers to the initial value. Voltage of the wind farm recovers to its initial value of almost 1 pu after almost 2 seconds and then it faces short length oscillations. Other machines stabilize in their initial value in the same time as the wind farm. In contrast to the wind farm the oscillation length of other machines is of an almost zero rate. Fig. 7 illustrates the voltage of the wind farm during the simulation (steady condition until 1 sec, fault condition until the stabilization along with some oscillation after 3 seconds of the total simulation). Fig. 8 depicts the voltage of all machines of the system during the simulation.

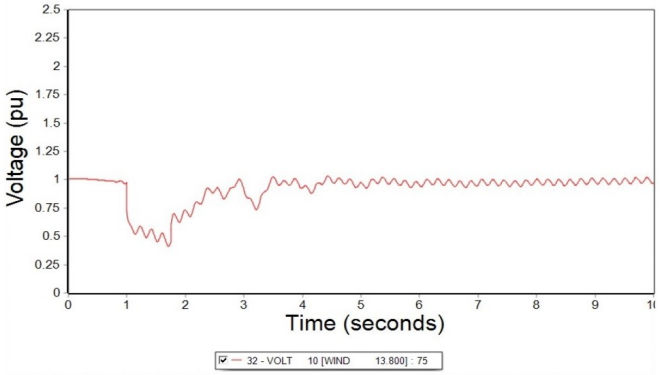


Fig. 7: Wind Farm Voltage.

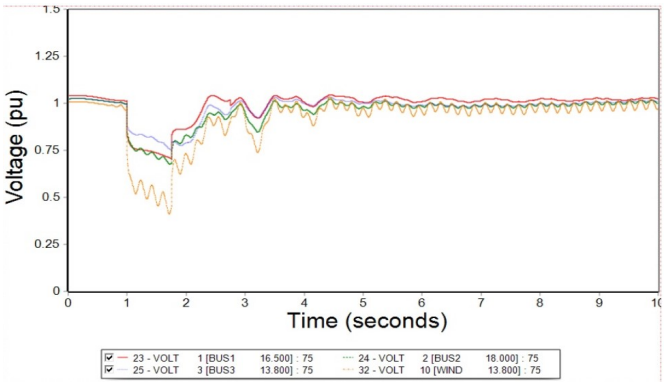


Fig. 8: Machines Voltage.

The correct operation of the network is ensured by the calculation of the critical fault clearing time (CFCT). The critical fault clearing time for this study was calculated at 0.75 sec (750 msec) as mentioned earlier. This critical time according to simulation results should not be exceeded in order for the system to be stable. In case the fault lasts longer, the system will be unstable. As a result of this, if the time duration of the fault (FD = 0.80 sec) is set to 0.80 sec (800 msec), the system becomes unstable even if the fault is cleared. Fig. 9 shows the active power of the machines of the system for fault duration 0.80 sec. It is observed that the width of oscillations which occur after the fault clearance, while the system is trying to retrieve a stable condition, is large. The smallest oscillation is observed at the wind farm. Active power of generator in bus 1 is observed to have the largest fluctuations. Reactive power of the system's machines presents great fluctuations as well. In Fig. 10 the oscillations that represent the reactive power of the machines of the system are depicted.

The voltages of the system at bus machines for fault duration greater than 0.75 seconds, becomes highly oscillated, as it is highlighted in Fig. 11, where the oscillations continue and are not reset to an acceptable value. In general, fault duration greater than 0.75 seconds should not be exceeded in order the system to stay stable.

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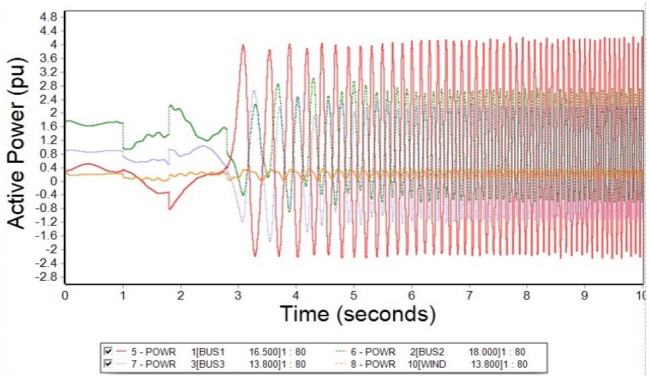


Fig. 9: Active power of the machines for FD = 0.80 sec.

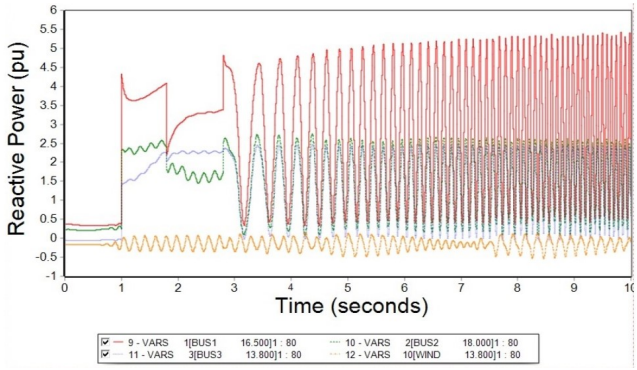


Fig. 10: Reactive power of the machine for for FD = 0.80 sec.

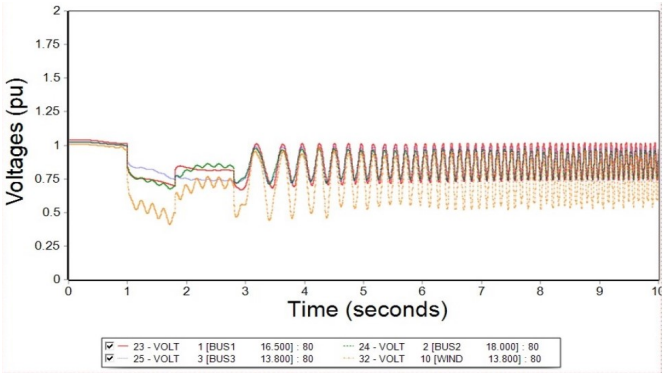


Fig. 11: Voltage of the bus machines for FD = 0.80 sec.

VI. Conclusion

In this research the integration of a DFIG wind farm in the IEEE 9-bus system was studied. A steady state and a dynamic analysis (transient stability analysis) was conducted and the study revealed that the critical fault clearing time is 0.75 sec for three-phase fault. This time should not be exceeded for the stability of the network. The results from the simulations showed that the 51 MW Wind Farm does not have a negative impact on network stability and operation. The active power, reactive power, voltage outputs of wind farm and for the machines of the network reset to their initial values after the fault clearing. It can easily be concluded that the integration of the wind farm into the power system is smooth except for some oscillations. Similar results, percentagewise, are observed in other studies [3], [9], [11]. Generally, the wind penetration into power systems and smarts grids spreads nowadays worldwide in a fast pace. The simulated transient results obtained in this research can offer future evidence for the integration of larger scale wind farms into networks.

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